

University of Moratuwa

Faculty of Engineering



GeoRemind

A Location-Aware Mobile Application for Smart Task and Errand
Reminders Using Geofencing

Project Proposal - Group 20

CS 4473- Mobile Computing

Team Members:

Senevirathne R.J.M. : 210598B

Balasooriya Y.B : 210058V

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1. Introduction

In today's fast-paced world, managing daily errands and personal to-do lists has become a significant challenge. Most people rely on standard reminder applications, but these tools are almost exclusively time-based. This creates a fundamental disconnect between the reminder and the context in which a task needs to be done. For example, a reminder to "Buy milk" set for 5:00 PM is useless if the user is stuck in traffic, far from a grocery store. This often leads to "notification fatigue," where users ignore or dismiss alerts that are not relevant to their immediate situation. This problem is amplified by the fact that users often remember tasks only when they are physically near the relevant location. A person might drive right past a pharmacy and only remember a needed prescription an hour later when they are already home. This inefficiency leads to wasted time, unnecessary travel, and the frustration of missed opportunities. Furthermore, any potential solution must address the specific challenges of the mobile environment in Sri Lanka. Constant background GPS tracking is not a viable option, as it rapidly drains a device's battery. An app that requires a persistent internet connection would be unreliable due to inconsistent network coverage and prohibitive for users conscious of mobile data costs. These issues highlight the need for a solution that is not only contextually aware but also lightweight, energy-efficient, and fully functional offline.

2. Proposed Methodology

To address the problems identified, we propose GeoRemind, a mobile application that makes task management intelligent and context-aware. GeoRemind uses core mobile computing features like geofencing, context filtering, and offline-first architecture to ensure that users receive the right reminder at the right place, not just only at the right time. The application workflow is designed for simplicity and efficiency. A user creates a task, such as "Withdraw cash," and attaches a location to it, like a nearby ATM, by selecting it on a map. The user can also define a radius for the alert (e.g., 100 meters). Once set, GeoRemind works silently in the background using the operating system's low-power location services. When the user physically enters the defined zone, they receive an interactive notification for their task. To make the system smarter and prevent false alarms, GeoRemind will introduce contextual integrity checks. The system will monitor if a user is simply passing by a location at high speed (e.g., in a car or bus) and will suppress the notification to avoid distractions. It can also be configured with a "dwell time" filter, where a reminder is only triggered if the user remains within the location's radius for a set period (e.g., two minutes), indicating a genuine opportunity to complete the task. At the end of a task, the user can mark it as complete directly from the notification. The app will maintain a simple history of completed tasks for user reference. Key features include:

- Location Awareness: To verify physical presence near a task location.
- Context Filtering: To reduce false alarms and improve notification relevance.
- Offline-First Design: To ensure core functionality works without an internet con-

nection.

Location checks will only be active when there are pending geofence reminders, ensuring minimal battery consumption when the app is idle. In this way, GeoRemind provides a reliable, efficient, and intelligent solution for managing real-world tasks.

2.1. System Architecture Design

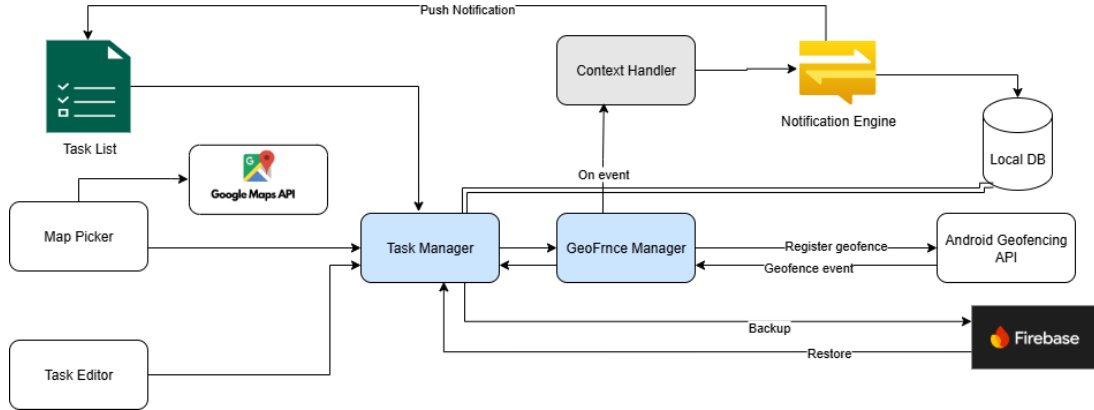


Figure 1: High-Level Architecture of the Proposed App

The system comprises the user’s mobile app (Flutter), a local on-device database for offline functionality (Room/SQLite), and the underlying Android OS services for location and notifications. An optional cloud backend can be used for backup and sync.

- Task & Geofence Manager (Mobile App) Creates and manages reminders. Defines the task description, location coordinates, geofence radius, and context rules (e.g., dwell time).
- Location & Context Engine (OS Services & App Logic) Uses Android’s native Geofencing API for low-power monitoring. The app’s logic applies context filters (speed, dwell time) to validated triggers from the OS before sending a notification.
- Local Database (Room/SQLite) Stores all user tasks and settings on the device, ensuring the app is fully functional offline and data persists across reboots.
- Optional Backend Services (Firebase) A lightweight Firebase backend can be used to support optional features like secure account creation and syncing reminders across multiple devices.

For this project, we will be using Flutter mobile app development and Room/SQLite as the primary database, with Firebase for optional cloud features.

2.2. Base Use Case

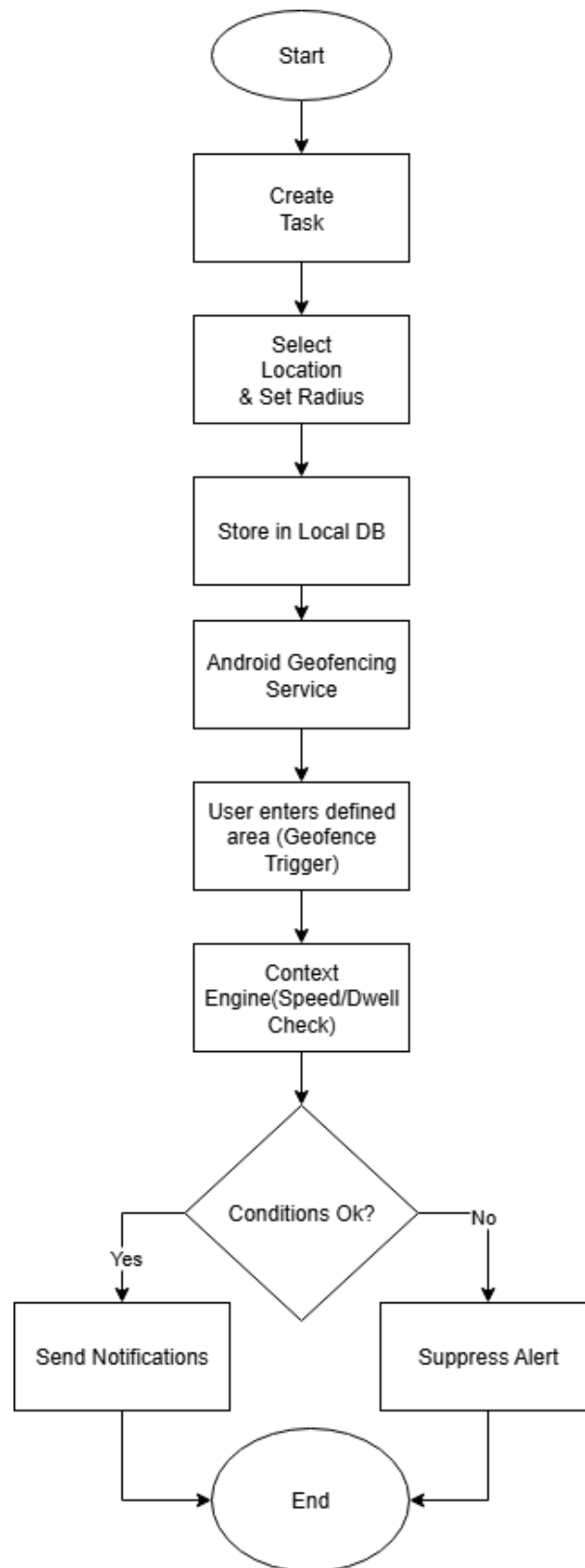


Figure 2: Base Use Case Diagram

3. Deliverables

The following deliverables will be implemented by the end of the project:

- **Mobile Application (Android, Flutter):** A primary user interface for creating, viewing, and managing tasks. An integrated map module for selecting task locations. A geofencing module for creating location-based triggers with custom radiuses. A smart notification system with context filters (speed and dwell time).
- **Local Data Persistence:** Full offline capability for all tasks using the Room/SQLite database.
- **Documentation:** how to use the app's features

4. Conclusion

GeoRemind will provide an intelligent, efficient, and reliable solution for everyday task management. By combining geofencing, context-awareness, and an offline-first design, it will ensure users receive relevant reminders precisely when and where they need them. This approach solves the core weaknesses of traditional time-based reminder apps while being perfectly optimized for the resource constraints of the modern mobile user.